

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457207.7538 +/- 0.002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.107 +/- 0.041
Transit depth (Rp/Rs)^2	1.15 +/- 0.87 [%]
Orbital Inclination (inc)	79.3 +/- 2.91
Airmass coefficient 1 (a1)	0.94 +/- 0.59
Airmass coefficient 2 (a2)	-0.14 +/- 0.86
Scatter in the residuals of the lightcurve fit is	68.77 %
Best Comparison Star	None
Optimal Aperture	2.29
Optimal Annulus	3.0
Transit Duration (day)	0.062 +/- 0.041

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.107 ± 0.041 vs 0.10538 (deviation 0.03σ)
Duration fit vs published	0.062 d vs 0.1167 d (deviation 0.95σ)
Mid-time O–C	4.5 min
Depth SNR	1.9
Residual scatter	68.77 %

Light curve

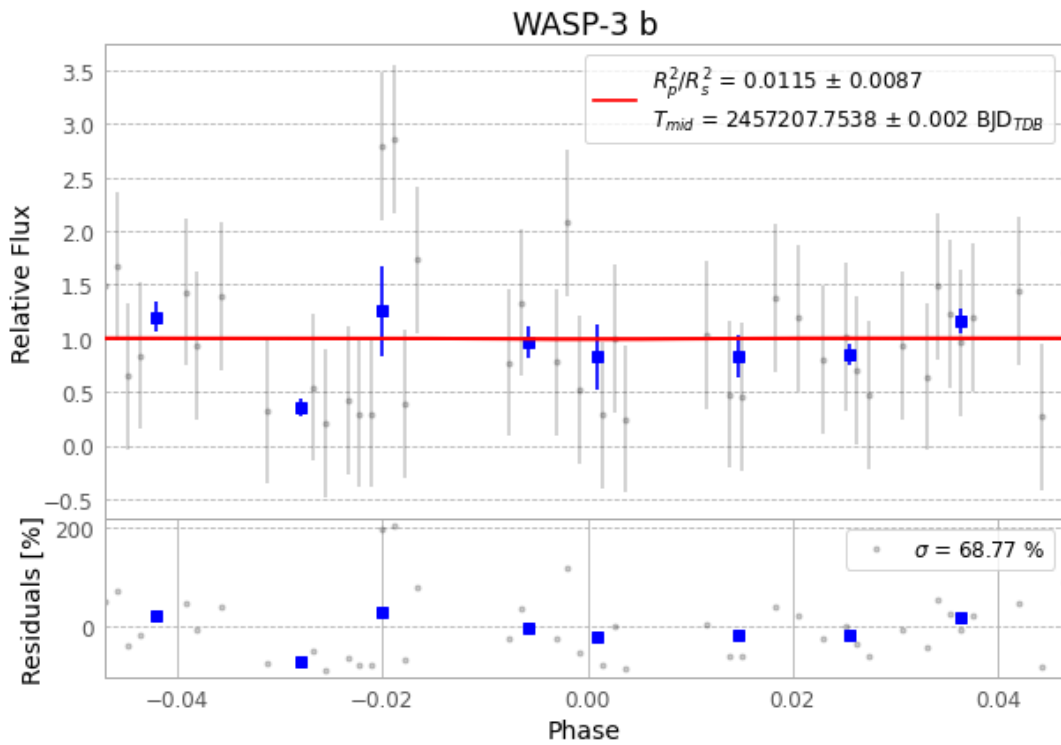


Figure 1 — FinalLightCurve_WASP-3 b_2015-07-03.png (SHA-256 63f5517b2aff8d01...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [394, 130], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d36e5ba199616b7f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

86 FITS frames (56 MB), WASP-3150704035412.FITS → WASP-3150704080912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-3-150704.log (912fb0306e5e...), FinalLightCurve_WASP-3 b_2015-07-03.png (63f5517b2aff...), FinalLightCurve_WASP-3 b_2015-07-03.csv (879ff415c4d3...)

Compute resources

Wall time 215.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 16:18Z.